AIM : Implementing Web Scraping in Python with BeautifulSoup.

Theory :

1. What is web scraping? Explain the role of the Requests library in web scraping.

Ans: Web scraping is the technique of automatically collecting data from websites using scripts or programs. It enables users to gather large volumes of information—such as text, images, and links—from web pages for analysis or other uses. In Python, the Requests library is essential for this process, as it sends HTTP requests to websites and retrieves their HTML content. It essentially acts as a connection between the user and the web server, allowing access to web pages similar to a browser. After obtaining the HTML content with Requests, it can be parsed and analyzed using libraries like BeautifulSoup.

2. What is BeautifulSoup? How does it help in parsing HTML?

Ans: BeautifulSoup is a Python library designed for parsing HTML and XML documents. It makes it easier to extract specific information from web pages by allowing users to navigate the HTML structure in a clear and simple way. The library transforms complex HTML code into a tree-like format, which helps in locating elements such as tags, classes, and IDs. Using its built-in methods, users can efficiently search, filter, and retrieve the required data. It is particularly helpful when working with poorly structured or messy HTML, as it can handle errors and simplify the overall parsing process.

3. Differentiate between web scraping and web crawling.

Ans: Web scraping and web crawling are closely related but serve different purposes. Web scraping focuses on extracting specific information from a particular website or web page, such as text, prices, or links. In contrast, web crawling involves automatically navigating through multiple web pages, often by following links from one page to another. Web crawlers are commonly used by search engines to index and organize content across the internet, while web scraping is mainly used to collect targeted data from selected sources. In simple terms, crawling is about finding and exploring web pages, whereas scraping is about retrieving useful data from them.

4. What are the commonly used methods in BeautifulSoup?

Ans: BeautifulSoup offers a variety of helpful methods for extracting and navigating data from HTML documents. Commonly used functions include find() and find_all(), which are used to locate specific tags or elements within the page. The get_text() method is used to extract the text content, while get() helps retrieve attribute values such as links or image sources. The select() method allows users to target elements using CSS selectors. Additionally, navigation features like .parent, .children, and

.next_sibling make it easier to move through the HTML structure. These tools make the process of data extraction both flexible and efficient.

5. What are the ethical and legal considerations in web scraping?

Ans: Web scraping should always be carried out responsibly by adhering to ethical and legal guidelines. Users should check a website's robots.txt file to determine which sections are allowed for access or scraping. It is also important to limit the number of requests sent in a short period to avoid overloading the server. Extracting copyrighted or sensitive information without permission can result in legal consequences. Moreover, personal or private data should not be collected without proper consent. Following a website's terms of service and using the collected data responsibly helps ensure ethical web scraping practices.

Install Required Library

Import Libraries

Send Request to website

Parse HTML content

Extract Book Titles

Extract Prices

Extract Rating

Create DataFrame

Save Data to CSV

Conclusion :

Web scraping was effectively carried out using Python libraries like Requests and BeautifulSoup. The Requests library was used to retrieve the HTML content from a website, while BeautifulSoup helped in parsing the content and extracting relevant information. Key details such as book titles, prices, and ratings were collected and organized into a structured format using a pandas DataFrame. The processed data was then stored in a CSV file for further analysis. This experiment shows how web scraping can be used efficiently to gather data from websites, while also emphasizing the importance of adhering to ethical and legal guidelines